BioSpecInfo

A guide for the people who use it — BioSpecInfo

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A guide for the people who use it — BioSpecInfo

This guide is written for people who study, not for people who program. You do not need to know anything about computing.

What it is

BioSpecInfo is an application for studying chemistry, biochemistry, pharmacology and astrochemistry. It opens like a website, but you can **install** it on your phone or computer and it works **without internet** too.

Three things to know straight away:

- **No sign-up needed.** No account, no email, no password.
- **It costs nothing.**
- **Your data stay yours.** Notes, files and conversations never leave your

device: there is no server they could reach.

👉 **samupropio1-ship-it.github.io/BioSpecInfo-v11**

Installing it (optional, but worth it)

Installing gives you an icon on your screen, full-screen opening and offline operation.

Device	How to do it
Android / Chrome	⋮ menu → *Install app* (or *Add to Home screen*)
iPhone / iPad	Share button → *Add to Home Screen*
Computer	Install icon in the address bar, on the right

After installing, open the app once with the network on: it will download everything it needs. From then on it works with no signal too.

Finding your way around

Element	Where it is	What it is for
🔍 **Search**	At the top	Searches the whole app. Shortcut: `Ctrl+K`
Section menu	Navigation bar	The 133 sections, grouped by subject
✨	Bottom right	Tools, settings and updates
Spectra	Bottom right	The assistant that answers questions

The things you will do most often

Analysing a molecule

1. Open the **Spectroscopy centre**
2. Type the molecule into the box. You can use:

- the **name** — `aspirin`, `caffeine`
- the **SMILES formula** — `CC(=O)Oc1ccccc1C(=O)O`
- or **draw it** with the editor

1. Press Enter

You get the 2D and 3D structure, the functional groups, the molecular weight, the predicted spectra (IR, NMR, mass, UV-Vis) and the pharmacological properties.

> **If you have no network:** searching by *name* does not work, because it has to consult an external database. The SMILES formula, on the other hand, always works.

Reading an IR spectrum

The spectrum is drawn as on a real instrument: the scale at the bottom runs from 4000 to 400 cm^{-1} (**from left to right the numbers go down**) and the bands point **downwards**.

The coloured area below 1500 cm^{-1} is the **fingerprint region**: that is where the specific molecule is recognised, not just its class. Below the chart you will find the list of bands with their assignment.

> The spectra are **predicted** from the structure, not measured in a laboratory. They are there to help you recognise functional groups and to understand the link between structure and spectrum. For experimental data, the app sends you to the official databases.

Everything about spectra, in one place

What you do with a spectrum used to live in **nine sections** across two different menus. Now there is 🔬 **Spectra centre**, a single section with ten tabs:

tab	what it is for
Read your spectrum	a file, a photo, a PDF
Predict NMR	^{1}H and ^{13}C assigned per atom
2D NMR and NOE	COSY, HSQC, HMBC
Mass	fragmentation and neutral losses
IR	bands and functional groups
UV-Vis	chromophores and λmax
Reference tables	^{1}H, ^{13}C, IR, J constants

tab	what it is for
Everything on one molecule	editor, spectra, groups, comparison
Elucidate a structure	from the data to a proposal
Practise	recognise the group from the spectrum

The panels are the same ones as before: they were **moved**, not copied — there is no second copy of anything. The old links still work: if something in the application used to take you to "Mass spectrum", it now opens the Spectra centre on the **Mass** tab.

> **Two tables of J constants, with different numbers.** There were two: one gave the aromatic ortho ^3J as 6–9 Hz, the other as 7–10. Whoever looked that number up got a different answer depending on where they came in. **One** remains, the one inside the Spectra centre, which sits in the section with its own reference list.

Changing language: the round button, top right

At the title's height, at the far right, there is a **round button** with the flag of the language in use. It opens into a **disc**: the fourteen flags on two shells — six inside, like a benzene ring, and eight outside — and at the centre appears the name of the language you are moving to and **how much it is translated**. That percentage is not hand-written: it is measured by applying the dictionary and counting what stays in Italian.

Touch a flag and the interface changes at once. `Esc` closes the disc.

> The 🌍 **Language** section remains, and does something else: it explains the languages, shows each one's coverage and says that Arabic carries the writing direction with it. That one **informs**, this one **acts**.

⚛️ Physics: eight computed sections

In the ⚛️ **Physics** category of the bar there are eight sections where the physics underneath chemistry is **computed**, not narrated. Every number you see is compared by the bench with an independent literature value.

section	what you do
🌊 **Schrödinger**	pick a potential (well, harmonic oscillator, Morse, double well) and see levels and wavefunctions with the exact solution beside them; launch a wave packet at a barrier (tunnelling); see the vibrations of HCl, H_2, CO and N_2 in cm^{-1}
🧮 **Hartree-Fock**	a real ab initio calculation (RHF/STO-3G) on eight molecules: energy, orbitals with their energies, Mulliken charges, a map of the chosen orbital, the dissociation curve
🎻 **Rovibrational**	the IR spectrum of a gaseous diatomic with its P and R branches: change temperature, resolution, isotope; rotational Raman of N_2, O_2, H_2, CO_2; from the lines to the bond length

section	what you do
🎲 **Statistical thermodynamics**	entropy and heat capacity of nine gases computed from molecular constants, contribution by contribution, and rotational and vibrational populations
🎱 **Molecular dynamics**	live Lennard-Jones particles: heat, compress and watch gas, liquid and crystal, with the speed distribution and g(r)
∑ **Mathematics for chemistry**	from FID to spectrum by Fourier transform; Hückel on twelve π systems; kinetics by Runge-Kutta; the Arrhenius line with errors from your data
💎 **Crystallography**	the 14 Bravais lattices to rotate, Miller planes, the powder pattern of eleven crystals with absences
☢️ **Radiochemistry**	decay chains (the hospitals' technetium generator), dating with ^{14}C, K-Ar and U-Pb, nuclear binding energy

📐 Mathematics: from basics to advanced

The 📐 **Mathematics** category holds eight sections that run on a real computer-algebra engine: type expressions as you would by hand (`2x² + sin(3x)` , `√(x+1)` , `e^(-k*t)` , `ln(x)/x`) and the results are **exact** — 1/3 stays 1/3. Antiderivatives appear only if differentiating them gives the original function back.

section	what you do
🔢 **Mathematics basics**	count significant figures and apply the measurement rules; logarithms and pH; solve equations with the working, from linear to polynomials (complex roots too) and the fractional equations of equilibrium (weak acid); practise with ever-new marked exercises
📈 **Function analysis**	type f(x): domain, symmetry, zeros, exact derivatives, maxima, minima, inflections, vertical, horizontal and oblique asymptotes, a moving tangent; and function families with a·f(b(x−c))+d
∫ **Derivatives, integrals and series**	the derivative with the list of rules used; the antiderivative and the definite integral with Riemann sums; the degree-n Taylor polynomial; left and right limits
🔲 **Linear algebra**	systems, determinant, inverse, rank, null space, eigenvalues, in exact fractions with every Gauss-Jordan step; plane transformations; balancing a reaction (redox with charges too) as the null space of the composition matrix
🌀 **Differential equations**	the direction field of y′ = f(x, y) with solutions starting where you tap; Euler, Heun and RK4 compared; the phase plane of predator-prey, pendulum, Van der Pol, Brusselator, SIR, with classified equilibria
🗻 **Functions of several variables**	3D surfaces to rotate and contour maps, gradient, Hessian, minima, maxima and saddles (the transition state), gradient descent; state functions, the Maxwell relation, exact differentials
ℂ **Complex numbers and Fourier**	complex numbers on the Argand plane, n-th roots, Euler's formula, Fourier series with the Gibbs phenomenon
📏 **Laboratory statistics**	mean, standard deviation, confidence interval; Grubbs and Dixon for outliers; t and F tests with p-values; uncertainty propagation; calibration line with LOD, LOQ and the sample uncertainty

🎓 UniBA exams: Chemistry, Mathematics and Physics, from basic to expert

🎓 **Chemistry exam** (🔬 Chemistry category) has 40 standard problems on four levels: basic (moles, stoichiometry, gases, solutions, thermochemistry, kinetics, equilibria, acids and bases, solubility, electrochemistry, the atom), intermediate (analytical: titrations, Beer-Lambert, gravimetry, EDTA, polyprotic acids), advanced (physical chemistry: van 't Hoff, Clausius-Clapeyron, polyenes, rotor and oscillator, Michaelis-Menten) and expert (Sackur-Tetrode, Debye-Hückel, Pourbaix). Reactions balance themselves. In Mathematics and Physics level **A** goes beyond the two courses: eigenvalues, ODE systems, Fourier, Laplace, triple integrals; waves, modern and quantum physics.

🎓 **Mathematics exam** (📐 category) and 🎓 **Physics exam** (⚛️ category) follow the syllabi of the Chemistry degree at the University of Bari. Each has the **syllabus** explained topic by topic with the method; the **solver**, which writes the step-by-step working with your data; **solved papers** in the style of the written exam (built on the syllabi: not the official papers). Mathematics also has a timed **exam simulation**. ⚛️ Physics also gains ⚡ **Electrodes and transition state** and 🌟 **Photochemistry, surfaces and polymers**.

🔭 Astrophysics, relativity, fluids, vector calculus, complex kinetics

In ⚛️ **Physics** there are 🔭 **Astrophysics**, ⏱️ **Special relativity** and 🌊 **Fluids**; in 📐 **Mathematics ∇ Vector calculus**; in ⚗️ **Physical Chemistry** 🌀 **Complex kinetics**. Under every animation there is a bar: ⏸/▶ pauses and resumes, ↺ restarts, **×¼ … ×4** changes speed, on the right the elapsed time. Animations stop by themselves when they leave the screen; with «reduce motion» on in the system they start paused and run with ▶.

🎯 The laboratories in motion

In the ⚛️ **Physics** category there are 🎯 **Mechanics in motion**, 〰️ **Waves and optics**, 🔌 **Circuits** and 🔥 **Thermodynamics in motion**; in 📐 **Mathematics** there are 🧮 **Numerical methods** and 📐 **Analytic geometry and vectors**; in ⚗️ **Physical Chemistry** there is ⚖️ **Equilibrium and Le Chatelier**. Each tab has sliders on the left and the simulation on the right, with the numbers describing it under the controls. In the equilibrium simulator the buttons add or remove a species, change volume or temperature: concentrations move until Q equals K again, and the note says which way the reaction goes.

🧊 3D Laboratory, charts and molecular views

In the ⚛️ **Physics** category the 🧊 **3D Laboratory** shows surfaces, atomic orbitals, motion in space, crystal structures and field lines in three dimensions. Rotate by dragging with a finger or the mouse and zoom with two fingers or the wheel. The buttons at the top right start auto-rotation (⟳), go back to the initial view (⌖), save an image (📷) and switch to full screen (⛶). Moving a finger or the mouse over a chart in any section shows a crosshair with the value of every curve at that point. Molecular 3D views have the same bar, plus the ✦ button choosing quality: automatic, high (with ambient occlusion), medium or low for slow phones. If «reduce motion» is on in the operating system there are no animations.

⚗️ ⚛️ 📐 The three macro-areas and the assistant

At the head of the 🔬 **Chemistry**, ⚛️ **Physics** and 📐 **Mathematics** categories is the area of that subject. In the 🤖 **Assistant** tab, write the problem as you would write it to your teacher, for example «a car at 90 km/h brakes at 5 m/s²», «how many grams of NaCl for 250 mL of 0.5 M solution» or «y'' + 4y = sin(x), y(0)=0, y'(0)=1», and press **Solve and explain**. The answer says what was understood; the **Data used** table shows in green the data read from the text, in yellow those inferred and in red those assumed, to be checked. If the chosen model is not the right one, the **Did you mean…?** buttons offer the others, and **Edit the data in the solver** opens the problem with its fields. The 🗂️ **All tools** tab gathers the subject's sections, with search; 🧭 **Levels** orders them from high school to PhD.

🧭 From high school to PhD

In the 📚 **Study** category, 🧭 **From high school to PhD** tells you, for chemistry, physics and mathematics, what you should be able to do at each level and opens the right tools with one click. The **Physics exam** now covers astrophysics, nuclear and quantum physics (up to variational and perturbation methods); the **Chemistry exam** organic chemistry, spectroscopy, biochemistry and a PhD level (Marcus, isotope effect, tunnelling); the **Mathematics exam** residues and the heat equation.

Where things are

The sections live in the **navigation bar**, grouped by subject, with 🛠️ **Utility** among them: laboratory, GHS, concept map, molecule comparison, molecule search, atlas, resources, materials, notes, file manager, statistics, data science, pomodoro, lab calculator, **articles**, and the two language ones.

The ✨ menu holds only what is *not* a section: the editors, the overlay labs, and actions such as exporting a figure, printing, sharing, updating. The rule is simple and always holds: **if something is in the bar, it is not also in the menu**. It used not to be, and the menu showed two "Data Science", two "File Manager", two "Articles" and two "Updates".

Two exceptions, and they have a reason:

- **Spectra** is not in the bar: it has its own dedicated button, the pulsing

one at the bottom. Three roads to the copilot were too many.

- The **spectrum reader** is not in the bar: it is a tab of the Spectra centre.

> **Two File Managers, and they are two different things.** In the bar, 📁 **File Manager** is for uploading and organising your own notes (PDF, images, Word). In the ✨ menu, 🔒 **Protected File Manager** is a separate page with the password, for the gallery and the custom sections. They had the same name and looked like a duplicate: now they are named for what they do.

The articles and the Nobel Prizes

📰 **Articles** (from the ✨ menu) has two tabs.

Latest articles shows what came out in the last two months, by topic: spectroscopy, synthesis, catalysis, drugs, materials, biochemistry, computational. It is not a hand-written list — that would be stale tomorrow: it comes from **Crossref**, the DOI registry, at the moment you open the tab. Each title leads to its **DOI**, which is a permanent address, and beside it there is a link to **read it in Italian**.

Nobel Prize in Chemistry lists every prize. Years, laureates and motivations come from the **Nobel Foundation**, live: they are not written inside the application. The **explanation** in Italian is — and it is matched to the *topic* of the official motivation, not to the year, so no attribution comes from here. For each prize you get the official page, the Foundation's scientific and popular material, the Italian translated view and an Italian Wikipedia search — all addresses **computed** from the year, not written.

> **If a prize has no explanation, I say so.** I would rather write "for this prize I have no explanation" and leave you the official motivation and the Foundation's material, which is the source, than make something up.

> **It needs the network.** Without it you do not see a stale list passed off as current: you see what I asked for, from whom, and why it did not arrive — with the address of the request, which you can open by hand.

Reading a spectrum you measured yourself

The **Spectrum reader** section does the opposite of the one above: it does not predict a spectrum from a structure, it reads one you already have.

There are **two buttons**, and they are not duplicates.

`📂 Open a file` filters nothing: whatever you have in your folder, open it. `📷 Photo or image of a spectrum` instead opens the **phone gallery** and also lets you take the picture there and then — on a phone these are two different system pickers, not two filters of the same one, and without the second there is no way to hand over a photograph.

A line under the buttons then **tells you what it decided the file is**:

What you open	What happens
a **PDF**, a **Word** file, an OpenDocument, a spreadsheet	it opens, shows you all of it and **works through** what it contains
a **photograph** (`.png` , `.jpg` , …)	both things: the **trace** is extracted and the image is also shown as a page, because it may be a photo of a spectrum or of a sheet
a **JCAMP-DX** (`.jdx` , `.dx`), compressed form included	it is a spectrum: the format itself says so
a **text** file or a `.csv`	decided **from the content**: if the numbers form a curve it is a spectrum, otherwise it is **problem text** and gets worked through

If you prefer, **paste** the data or the problem text into the box and press **Read and work it out**: the same reasoning applies.

> There used to be two cards, and the spectrum one allowed only `.jdx` and `.csv`. On a phone that filter did not narrow, it **hid**: anyone arriving with the PDF of an exam saw an empty folder. Now there is one door and it filters nothing.

If you have no file at hand, the three buttons **IR example**, **MS example** and **NMR example** load a sample spectrum, and **example: an exam problem** loads a complete problem.

Inside the section there is 📖 **How to use it — full guide**: seven short chapters covering the two buttons step by step, the calibration, what the program does by itself, what it cannot do, how to recognise a wrong result from the symptom you see, and how to take the numbers away.

A spectrum you only have as a figure

The trace is extracted from the **pixels**, and the chart appears at once. Then it must be **calibrated**, because the axis numbers are not in the image:

field	what you write in it
x at left / x at right	the values at the two edges of the plot — for an IR typically 4000 and 400 **in that order**, because an IR is written with decreasing wavenumbers
y at bottom / y at top	the values of the vertical axis; for a figure in %T, 0 at the bottom and 100 at the top
y unit	`%T`, `T`, `A` or intensity. Picking `%T` derives the absorbance with **$A = -\log_{10} T$**, which is the definition
x/y logarithmic	if an axis is logarithmic, say so: applying a linear map to it does not raise an error, it produces wrong numbers that look right
frame	normally I find it; "the whole figure" is for when the image is already cropped to the plot only

Change a field and the chart redraws by itself: there is nothing to confirm. Below it the **diagnostics** appear — how many columns, whether the frame was found and on which sides, background and threshold, how many columns were grid — because a number without its how cannot be challenged.

What I do by myself: I find the **axis frame** and digitise only what is inside it (the numbers written on the axes are dark pixels just like the trace); I **survive the grid**, picking in each column the run near where the trace was a pixel earlier and dark enough to be the trace; I read at **sub-pixel**, because the centre of a two-pixel line sits in between.

> **How much I get wrong.** Measured over seven constructed figures whose band centres are known: **mean 2.2 cm^{-1}** deviation, **worst 7.3**. One column of a figure 900 pixels wide over 3600 cm^{-1} is worth about 4 cm^{-1}: below that there is nothing to gain, it is the limit of the figure.

> **If the calibration is wrong**, the peaks come out at wrong numbers while sitting in the right place on the figure: the shape is recovered from the pixels, the calibration is not. And a photo taken at an angle carries a distortion I do not straighten — frame the figure head-on.

With **download CSV** and **download JCAMP-DX** the numbers leave this place: any spectroscopy program opens the JCAMP, and this very reader reads it back identical. Inside it is written that the calibration was given by hand.

Below the chart you will find the peaks found with their signal-to-noise ratio, and for each band **all** compatible assignments. If you read two rows at 1715 cm^{-1}, that is not a defect: at that wavenumber a ketone and a carboxylic acid are both possible, and naming only one would be inventing.

> The reader **does not tell you which molecule it is**. A set of bands is compatible with many different molecules: the step from spectrum to structure is yours to take, with everything else you know about the sample.

Opening a problem and having it worked through

The same button, `📂 Open a file`. It takes the PDF of the exam, a photograph of the sheet, a Word file, plain text — and shows you **everything** inside: every page drawn (click to enlarge) and the full text.

Then, if the text contains spectroscopic data, it recognises them and puts them in a table **with where it took each one from**. That table sits above the working on purpose: look at it first. If it mistook a mass for an IR band, the working that follows will be impeccable and the answer wrong, and you catch it there.

The working lists the steps with the **evidence** for each and how certain it is: degrees of unsaturation are arithmetic, the carbon count from the M+1 peak is an estimate with a margin, an IR band is *compatible with* several things.

> **It does not give you the structure.** A set of spectroscopic data is compatible with more than one molecule, and a program producing a single one would offer a certainty the data do not contain. What you can do is the opposite: write a structure yourself and the program **compares** it, signal by signal, saying where it fails.

> **If the document is a scan**, the text cannot be read: a scanned page contains pixels, not letters, and there is no optical character recognition. The app tells you instead of pretending the sheet was empty. The pages stay visible, and if one is a photograph of a spectrum the trace can be extracted with **Open an image**.

Choosing the language

The **Languages** section holds **fourteen** languages, with a search field to find them quickly: Italian, English, Spanish, French, German, Portuguese, Dutch, Polish, Romanian, Greek, Russian, Chinese, Japanese and Arabic. Choosing Arabic also flips the page's **direction**, right to left.

Next to it is **Chemical languages**, which is a different thing: it converts a molecule among the twenty-five notations chemistry writes it in — SMILES, InChI, SMARTS, molfile, XYZ, PDB, CML, the fingerprints — and, for a declared class of simple molecules, writes its **IUPAC name**. Outside that class it says so, rather than attempting a wrong name.

Studying with cards

1. Open the study section
2. Generate a deck of cards
3. For each card, after answering, say how well you remembered it

The app works out **when** to show it to you again: the things you know well come back rarely, the uncertain ones come back soon. When you reopen, you only find the cards that are due.

Looking up a drug

Open **Pharmacology**. For each of the 263 drugs you will find the structure, molecular weight, mechanism of action, indications, adverse effects and class.

> ⚕ **For teaching purposes only.** This information does not replace the judgement of a doctor or a pharmacist, and is not there to decide a therapy.

Spectra, the assistant

Spectra answers chemistry questions and can use the app's own tools: calculate, generate spectra, look up constants, balance reactions.

Getting it working

Spectra relies on an external artificial-intelligence service, and you need a free **key**. Two routes:

The simplest — a key of your own

1. Go to console.groq.com and create a key

(free, it takes two minutes)

1. Open Spectra, paste the key into the 🔑 box
2. Done

The definitive one — a proxy

If services often fail to answer for you, or you would rather not keep the key on your phone, you can publish a *proxy*: a small free program that keeps the key safe and makes even the services that are blocked from the browser work.

In the assistant's **Proxy** box you will find **"How to set it up, in 4 steps"**, with the commands ready to copy.

When Spectra does not answer

Almost always it is not the app's fault but the external service's. Press 🔌 **Test** in Spectra's panel: in a few seconds it tells you **which services your device can actually reach** — and you need no key to find out.

If a service stops answering, Spectra remembers it for 24 hours and moves it to the bottom of the list, under *"Did not answer from this device"*. It does not delete it: tomorrow it might work again.

The File Manager

A personal archive for photos, documents and HTML pages, protected by a password.

> ⚠️ **Something you need to know.** This protection is a **deterrent**, not real security: whoever knows where to look can get around it. Do not put confidential documents in it.

The files stay on your device. **There is no copy anywhere else:** if you clear your browser data or change phone, they are gone. Use the export for the things you care about.

Frequently asked questions

Do I have to pay for anything? No. The app is free. If you use Spectra, the AI service's free key has daily limits, but nothing is paid.

Does it really work without internet? Yes, after the first opening. The only things left out are those that by their nature have to ask someone else: searching molecules by name, 3D structures downloaded on the spot, and Spectra.

Are my notes saved somewhere? Only on your device. There is no server: nobody can read them, but nobody can recover them for you either if you lose them.

How do I delete my data? ✨ menu → Settings → delete data. You can choose **what**: only the keys, only the conversations, only the study data, or everything.

I updated but I do not see anything new. ✨ menu → **Updates** → *Check*. There you can also see which version you are running.

Can I trust the spectra for a report? For recognising functional groups and understanding the structure-spectrum relationship, yes. For a figure to be cited as experimental, no: they are predictions. The app sends you to the official databases (NIST, SDBS) with direct links.

I found an error in a chemical datum. Report it: open an *issue* on the GitHub repository. The data go through automatic checks, but no check finds everything.

If something goes wrong

Problem	What to try
The app does not open	Reload it. If it persists, ✨ menu → Updates → Check
I do not see the new features	The version is in the ✨ menu → Updates
Spectra does not answer	The 🔌 **Test** button: it tells you which services you reach
"Out of space"	✨ menu → Settings → delete the old conversations
A section stays empty	Check the network: some sections load data from the internet
The charts are blurred	Reload the page; the charts are redrawn at your screen's resolution

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Contacts

Author: Samuele Pio Provenzano **Repository and reports:** github.com/samupropio1-ship-it/BioSpecInfo-v11

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